

Introducing Robots in a Warehouse: Lessons Learned from a Field Study at a Supermarket Chain

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Abstract

This paper presents a field study conducted in a distribution warehouse of a supermarket chain, Colruyt Group, in Belgium. It focuses on workers' acceptance of autonomous mobile robots (AMRs) with anthropomorphic features. The robots were developed by an on-site technical team, and our research team of social scientists was consulted to explore workers' perspectives. We compared four versions of the robot (without any features, with a name, with eyes and with a display) using mixed methods, including ethnography, interviews and surveys. Collecting data via periodic surveys proved challenging since the warehouse employees were busy during peak seasons (high workload) and due to technical issues with the robots. These factors necessitated a more flexible approach. Nonetheless, our study provided valuable insights that were considered during the robot implementation. In this case study, we summarise the results, lessons learned, and suggestions that could help navigate future field studies in industrial environments.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Field studies.

Keywords

Autonomous mobile robots, Warehouse, Workers' involvement, Workers' acceptance

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1 INTRODUCTION AND BACKGROUND

Existing studies listing lessons learned from user involvement within HRI focus mainly on social and service robots [2, 13, 16], safety and legal requirements [19] and technical requirements [7] while performing experiments. Limited research [3] explores long-term acceptance and overall experience during field study and collaboration with industry partners. In particular, robot acceptance in warehouses tends to be overlooked. According to Jacob et al.'s systematic review [8], only a few field studies focus on the acceptance of workers in warehouse environments. Furthermore, only eight out of 36 studies investigated autonomous mobile robots (AMR).

AMRs are robots that function autonomously and do not need designated routes or tracks, thanks to sensors that can detect and avoid obstacles [23]. They are specifically designed to adapt to the environment and can, therefore, offer high flexibility whilst requiring minimal changes in the warehouse logistics [8].

This paper presents a mixed methods approach to evaluating users' acceptance of autonomous mobile robots (AMR) by workers in a warehouse dedicated to distributing goods to the grocery store chain Colruyt Group in Belgium. Specifically, we followed a trajectory of an AMR, which was designed and developed in-house and is meant for transferring palettes with goods safely and correctly from point A (reception) to B (aisle), as seen in Step 2 in Figure 1 in Chapter 2. We worked closely with a technical team and were in contact with workers at multiple time points within one year. To begin with, we observed current processes to improve our understanding of the context in which the robots will be implemented. Next, we conducted interviews with several workers ($n = 4$). After that, we set up a user acceptance survey at four different time points during the design process when various humanised features were gradually implemented together with additional observations ($n = 3$). As previous studies showed, humanised features on robots, such as eyes, can help people interact with them and perceive them as more trustworthy [5, 10], positively affecting users' acceptance. Moreover, basic social interactions, like greetings, can help with users' perception of the robot and facilitate the human-robot interaction [12, 21].



Figure 1: Current process in the warehouse: 1) Receiving goods from trucks, 2) Moving pallets to the right aisle, 3) Lifting the goods to the right shelf, 4) Picking the goods to a truck going to the supermarket

The research in the case study focused on the following research questions:

- What is the employee's experience regarding the AMRs?
- Which features do the AMRs need for a smooth, safe and accepted human-robot interaction?

The study brought multiple challenges, which affected the results. However, we suggest that sharing our experiences within the CHI community, best practices and suggestions for future studies can help not only researchers but also companies that would like to implement new technology that affects workers' tasks. Our contribution is, therefore, three-fold: Firstly, we present results from the case study about the acceptance of AMRs. Secondly, we reflect on lessons learned coming from the research that affected the results. Thirdly, we make a list of suggestions for future field studies in industry.

2 CASE STUDY AT COLRUYT WAREHOUSE

2.1 Context and Current Logistics at the Warehouse

Colruyt Group is a Belgian supermarket retail chain currently employing more than 33,000 employees [4]. Our case study focused on one of their distribution centres located in the Flemish part of the country. Despite its location in Flanders, where Dutch is the main spoken language, the company operates in two official languages. In practice, most blue-collar workers speak French.

The main logistic process in this warehouse is the following (Figure 1):

- The first point at the distribution centre is the area of a receptionist. This is the place where trucks with the goods arrive. The receptionists receive the goods from the drivers. The truck drivers unload the goods and place the pallets on the ground at the specific marked zone. The receptionist is in contact with the outside world and the drivers. The receptionists mark the goods with stickers indicating a picking position where the goods get picked.
- Workers A ride the pallet trucks, pick up pallets with stickers, and drive them to the right aisle, playing a crucial role in warehouse logistics.

- Workers B, who drive forklifts, arrive and put the pallets on the right shelves and later take them down again to put them on the picking positions on floor level.
- Workers C drive another cart type to pick up the goods and products that need to be transported to the store.

2.2 The New Process with Autonomous Mobile Robots

The AMR should support worker A in Step 2, who is picking up the pallets from the receptionist. The motivations behind including the robots in the workspace stated by Colruyt Group were: 1) the current lack of workers and difficulty finding new employees due to the labour market shortage, 2) no added value of human workers transporting pallets, 3) the potential for deploying workers where they are most valuable, 4) cutting back on logistics costs.

As the company could not find robots that would fit their needs (the space in the warehouse is limited and too narrow for commercially available robots), they decided to develop the robot via the in-house development team with external design agency consultations. The robots navigate within the environment via LIDAR (Laser Imaging Detection and Ranging) and SLAM (Simultaneous Localization and Mapping) technology [22]. The software is programmed to match the traffic rules applied in the warehouse. The idea was to gradually add different humanised features (T1: added a name to the robot, T2: added a name and a display with eyes to the robot, T3: added a name and a display with text to the robot; see Figure 2) and evaluate which type of AMRs is most accepted by workers. The Colruyt Group selected the features and their order based on the technical feasibility. The name remained in all 3 data points, as it was decided by Colruyt Group to be used for the identification of a specific robot. Our research team was invited to assess the acceptance per each type of AMR model. During the study, two robots were operating on site, in the future, it is planned to integrate 7-8 of them. When operating, in this stage, robots were followed by the engineers to ensure safety.

3 METHODOLOGY

Data collection started in December 2022, when the first observation was made, followed by interviews ($n = 4$) in August 2023 to understand the context and motivations behind the implementation. These initial methods were mainly used to understand the context

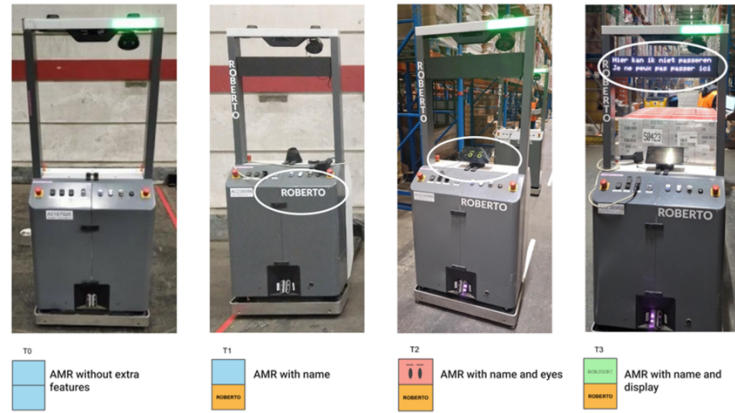


Figure 2: Overview of the design versions of the robot

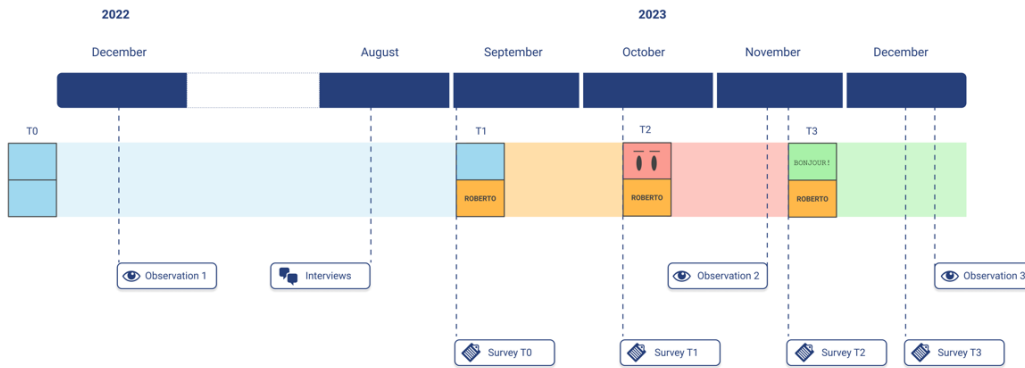


Figure 3: Research design timeline

[9] and to develop the surveys. To structure the observation notes, we followed the dimensions (space, activity, object, act, event, time, goal and feeling) indicated by Spradley [14]. Surveys were filled out by workers ($n = 37\text{--}59$) at 4 different time points based on the features added (between September 2023, when the prototype was first upgraded, until December 2023). For details, see Figure 3. The survey data collection started once the robots were integrated into the shifts so workers could comment on their actual experience. During the study, we also kept a research diary with reflections and notes about data collection, changes in the research plan and immediate thoughts and observations after data entry.

3.1 Participants

The participants were selected by the management team of the Colruyt Group. An overview of the participants can be found in Table 1 below.

The survey was available in Dutch and French because these were the main languages spoken in the warehouse. The language was also checked for the level of understandability. It was designed to correspond to the B1 (intermediate) CEFR level (Common European Framework of Reference for Languages), as it is a minimum

requirement for this position. The same workers who were selected for the first data collection point were also invited to all subsequent ones. However, the participation remained voluntary, and the number of participants tended to decline, with varying attendance across the data collection points. 41% ($n = 59$) of workers did participate in all time points, 37% three times, 19% twice, and only 3% dropped out after the first round. Reasons for not participating differed and included holidays, sick leaves, or no interest in participating anymore. The mean of workers' age who completed the survey was 42 years ($s.d. = \pm 2.6$), with 83% men and 17% women.

The study was approved by the Ethical Board of the Vrije Universiteit of Brussels. Informed consent was provided to all participants prior to the data collection.

3.2 Survey and Data Collection

While we initially proposed a more participatory approach, such as small focus groups, due to the limited availability of workers during their shifts and to involve as many workers as possible, we opted for a quantitative measurement, which was aligned with the development team's preference.

Table 1: Overview of participants

	T0	T1	T2	T3
French speaking	52	42	34	38
Dutch speaking	7	4	3	5
Total n of participants	59	46	37	43

The main measurements of the acceptance were adapted from existing (well-known and used) 5-point scales, focusing on intention to use [18] (e.g., *I plan to work with AMRs in the coming months*), performance expectancy (e.g., *AMRs seem useful for Colruyt*), trust [20] (e.g., *I understand the capabilities of the AMR*), safety [11] (e.g., *The AMR will prioritise the safety of the employees*), job insecurity [6] (e.g., *I am sure I will be able to keep my job despite using AMRs*), social influence [18] (e.g., *People who are important to me think that I should use AMRs*) [6] and anthropomorphism (e.g., *Fake-natural, Artificial-Lifelike*) [1]. Open questions, such as: *What do you think about the AMRs with [feature x]?* and *What still needs to be improved about the AMRs with [feature x]?* complemented the survey.

Most of the questions regarding acceptance were consistent across all the data collection time points. However, some performance expectancy questions were removed at T1, as we did not perceive that adding a name would influence certain items (e.g., *I understand what the AMR should do*). Questions related to social and demographic characteristics, job security and social influence were asked only at T0, as no changes were expected over time.

The total number of questions differed from 24-46 based on the time point. The complete list of questions is provided as supplementary material. The surveys were printed and filled out manually.

The first data collection (T0) was conducted in person, with two researchers distributing the survey to both morning and afternoon shifts at the end of their regular briefing meetings. However, it was concluded by the management that it was too time-consuming, so the following data collection slots were suggested rather in the early morning before the shift started. As it was not possible to reach the warehouse at this time by public transport, we prepared the surveys in advance, let the management distribute them to the workers, and later collected the completed surveys. The handing over and collecting of the surveys were also used for an hour-long observation of the robots during the shifts.

3.3 Data Analysis

After the data collection, surveys were manually entered into the QualtricsXM software. The exported quantitative results were analysed in SPSS and R software, and open questions were exported and coded in MAXQDA2022. The interim results were communicated in detail to Colruyt's development team after each data collection time point. To check for significant differences between the scales in all time points, we used the Kruskal-Wallis test. In the following section, we summarise the main findings and list the lessons learned from the case study.

4 MAIN RESULTS

Firstly, we will present the comments about design and functionality from both closed and open questions in the survey. Secondly, we

will summarise the quantitative results of the surveys. Thirdly, we will describe additional feedback found in the surveys, followed by limitations and lessons learned.

4.1 Design and Functionality

4.1.1 T0: AMR without humanised features. The initial survey served as a benchmark. There were not many comments about the design of AMR. However, a group of enthusiasts (26%, $n = 59$) expressed their excitement about the implementation and the project. Comments about improvement in open questions referred mainly to the functionality, such as limited speed, slowing work down or stopping too often, for example: "I think it's not fully operational yet because it's too slow ... and it often blocks the order pickers in the aisles."

4.1.2 T1: AMR with name. The name was seen by workers mainly from a practical perspective, and many of them saw an advantage in making the identification easier. 20% ($n = 46$) found naming the robots a nice/funny idea. During observations, we also noticed that some workers called the robots by their names (Roberto and Daniel) and said "hello" out loud to them. In the survey, suggestions were made to add a female version or other, more ordinary names. The number of remarks regarding the functionality aspect of the robot decreased, even though there were some remarks about the need for higher speed and smoother movement.

4.1.3 T2: AMR with eyes. Data collection for the third time point was postponed several times (making the difference between the time points longer than anticipated) due to technical adjustments related to the added eyes. In addition, the AMRs were not operating as frequently as initially planned, so we had to adjust the timeline to ensure workers had the same level of exposure as the previous robot versions.

For these reasons and also given the hectic period of the year (incoming Advent and Christmas season in the stores), the T2 survey had the lowest participation of workers ($n = 37$). Based on the previous research [5, 10], there were expectations that the eyes would make a difference. However, there was no significant difference found between the other versions. Moreover, there were only limited comments about the eyes found in the open questions, and 14% of the participants did not notice or see the eyes at all. There were some comments about liking the eyes (22%) but also remarks that they are not useful. Comments about functionality, such as stopping too often and the need for better fluidity of the robot, emerged again: "... these machines are already perfect, but we just need to improve the fact that they sometimes stop on their own in the aisles and a technician has to come and start them up."

Also, there were some comments about wanting the robot's eyes to blink, which they already do, but apparently, it was not noticeable enough to the workers.

4.1.4 T3: AMR with LED display. As participation had a declining tendency, we planned to come again in person for the data collection. However, due to a miscommunication, the managers collected the data, finding the surveys at the site and distributing them to the workers earlier than anticipated, which made the timeframe between the two timepoints shorter than planned. Compared to T2, 30% of participants thought that display could help with communication. These workers appreciated the idea of getting more information via display about the robot's actions but they would prefer more concrete information, such as "I will stop now". The speed at this point was almost the same as the carts driven by workers. Thus, there were almost no comments about speed. However, there were new comments about the need for AMR to give priority and that robots do not stop when they should.

4.2 Performance Expectancy, Trust and Intention to Use

4.2.1 Trust in AMR. Regarding trust, there was minimal variation across the time points. However, the AMR with a display (T3) performed significantly worse than the other versions. For example, 40% (n = 42) of workers in T3 disagreed with the statement "*The AMR will not make a mistake*", compared to 29% in T0 (n = 55), 37% in T1 (n = 43), and 27% in T2 (n = 34). Furthermore, workers perceived the T3 version as less consistent and reported feeling less comfortable relying on the information provided by the robot, particularly when compared to the T2 version with eyes.

4.2.2 Performance expectancy. No significant differences were observed in the perceived usefulness of AMRs across the time points. Many workers consistently viewed AMRs as useful for Colruyt: T0 (53%; n = 59), T1 (59%, n = 44), T2 (57%, n = 35), and T3 (55%, n = 42). However, fewer participants believed that AMRs would notably accelerate the speed of tasks. For instance, agreement with this statement was 25% in T0 (n = 57), 39% in T1 (n = 44), and 34% in T2 (n = 35).

4.2.3 Safety. Also, in safety, no significant difference between time points was found. However, AMR with a display (T3) performed worse than the other AMR versions. For example, 65% (n = 43) of workers in T3 agreed with the statement "*AMR will stop in time*", compared to 72% in T0 (n = 56) and 67% in T2 (n = 36).

4.2.4 Intention to use AMR. Participants were less willing to use AMR with display (T3) than the other versions. In T3, 47% (n = 42) of participants agreed with the statement "*I intend to work with the AMR in the coming months*," compared to T0 (51%, n = 57), T1 (62%, n = 42), and T2 (61%, n = 36). However, no significant differences were found between the two specific versions, only the overall difference. Some surveys included written comments stating that workers have no choice whether they will be working with the robot, as it does not depend on the workers themselves.

4.2.5 Anthropomorphic features. Neither anthropomorphic scale showed any significant difference between time points. Surprisingly, AMR with eyes (T2) performed the worst out of all time points and was seen as the least human-like.

4.3 Remarks and Workers' Feedback

4.3.1 Clarity and communication. At every time point of the data collection, participants had additional questions regarding the robots, especially about the changes in the warehouse linked to the robot implementation, such as: *How many AMRs will operate? How will the current task change?* Also, how AMRs were operating was not self-explanatory enough, for example, the eyes' function (changing based on the actions that AMRs were doing) was barely noticeable. There were also expectations and misunderstandings about how AMRs actually work or will work.

These questions were communicated to the development team and passed on to the communication team to develop a communication strategy. However, the strategy only took place after our third data collection. Hence, the workers' questions were repeated in the surveys. The insights also point to the need for the robots to be more self-explanatory and more explainable by design to the workers.

4.3.2 Feedback from participants. During the first data collection T0, where the research team was present, participants showed genuine interest in the research and their participation. As we were not present to collect the other data collection, we also decided to ask them about their experience with being involved in the last survey (T3) to get feedback. 73% (n = 41) of participants liked being involved in the study, and 58% (n = 41) would like to be involved in similar future studies to give feedback about technology implementation.

4.4 Limitations

The limitations accompanying this study clearly affected the results listed above. Firstly, the AMRs were not actively operating as planned, sometimes, running only for limited periods, which could result in overlooking the changes in the design. In particular, AMRs with eyes and AMRs with display were often overlooked by workers, as they rarely saw them operating. Additionally, there was a slight decline in the number of participants, mainly caused by holidays, sick leaves and busy times in the warehouse. These reasons also caused the delay in data collection multiple times. As a result, operating times of the robots varied, and frequent pauses may have caused workers to forget their experiences with the AMRs. The busy season also left workers with limited time to complete the survey. In the following section, we will outline the lessons learned from our experience.

4.5 Lessons learned

4.5.1 Observations and field work necessary to understand the context and create the measurements. It was crucial to include observations and interviews to create the survey and to understand the context in which the workers and AMRs were working. Without this, it would not have been possible to create the survey, and we would also not be able to interpret and understand workers' comments in the open questions.

4.5.2 Implication of the survey as a method. The Bortnick et al.'s frequently used anthropomorphism scale [1] was not clear to all participants, despite adding an example to improve understandability. This opens a discussion about scales that are widely used but might not be as understandable for all types of participants.

Workers also showed fatigue with repetitive questions, seen in declining participation, skipped questions/whole pages, or repeated answers for multiple questions. Moreover, questions regarding the intention to use were sometimes commented on by a written note such as: "Do I have a choice?", which reflect the reality that, as the technology is introduced, workers will be encouraged to collaborate with the robots, with limited flexibility in opting out of their use.

4.5.3 Busy period, busy environment. Even though we aimed to make the survey as concise as possible, it still proved demanding for the gatekeepers and did not align well with their busy schedules, leading to multiple delays in data collection. The first survey completion took approximately 15 to 20 minutes, compared to about 7 minutes for the test survey completed by the research team.

4.5.4 Clear communication with workers within the workplace was needed. The information about the changes in the environment regarding the robot implementation needed to be communicated more quickly and efficiently to the workers by the management, as related questions from workers appeared at all time points, as the communication strategy was delayed.

4.5.5 Technical and design challenges impacting research. As the robot was being developed during our research, there were multiple times when technical issues influenced our data collection: for example, the AMRs were not operating as planned during our observation, or they were not operating at the time planned, so workers did not see them in action. This influenced the research plan, such as multiple data collection delay and therefore influenced the results and reproducibility.

Changes in the robot's design also affected the results. Except for adding the name, the addition of the eyes and display was not visible enough to the workers to comment on the design. Many of the participants did not notice any change, or it did not make any difference to them.

4.5.6 Outsourcing the data collection added to the limitations. Being physically present during the data collection had an advantage, as we could directly help workers if some questions were not clear. Moreover, we had a chance to explain our research and connect with the participants. From a practical point of view, it was easier to let managers collect the data. However, we did not have an oversight of the survey distribution, which, we assume could benefit the data quality and result in a higher number of participants.

5 DISCUSSION AND IMPLICATION FOR FUTURE

User involvement has been proved as a precondition for successful implementation and acceptance [15, 17]. Nevertheless, using the bottom-up approach in the real work context can be surrounded by unexpected events that can influence the implementation process, such as higher workload, technical issues, changes in management and other challenges.

The study invites the CHI community to reflect on the problem of the importance for users' involvement, along with the related implications that need to be addressed. Even though the company expressed its interest in users' involvement and was happy to collaborate with academic researchers, the workers' involvement early in the design process was not possible and collecting feedback only during implementation was preferred due to less extensive time investment.

The outcome, however, showed that workers wanted to be heard and liked the idea of expressing their own opinion, which was especially remarkable when we were present during the data collection and we could observe their genuine interest. Issues connected with development and technical functionalities also influenced workers' perspectives. While the main focus of the study was different designs, workers constantly commented on the speed, efficiency, reliability and safety of the AMRs rather than the design itself, which was secondary in their view. Despite the challenges, the insights gained were shared with the development team and management, guiding the implementation, further development of the robots, and the selection of the most suitable AMR version.

Lessons learned showed us that conducting research in an industry workplace requires flexibility, therefore an adaptable research design, and perseverance. There is also room for discussion about the best trade-off between the time efficiency of the method and the validity of the method. The survey exhaustion was evident after the second round and questions often remained not answered. Nonetheless, these should not limit or discourage researchers or companies from proceeding with field research, which can bring many benefits and prevent future issues. This study provides valuable insights that could be considered when implementing human-centred automation solutions in industrial settings. Building on these findings, we propose the following suggestions for future field studies within the industrial workplace:

- **Start early on:** Co-creation methods and users' involvement in the early stage can prevent design flaws. In this case study, we could avoid implementing the eyes, which were barely visible and not noticeable to the workers. However, keep in mind that these methods require a significant amount of the workers' time, and it might not always be possible to apply them in a dynamic workplace environment (as it was in our case).
- **Be flexible and make an adaptable research design:** Choose flexible methods so the outcomes are not greatly affected when conditions change, which can happen quickly and often in an industry environment.
- **Keep a research diary:** The immediate reflections after data collection, as well as the reasons for delays or changes made, are easy to forget. Keeping a research diary helps to keep track on all the research activities and helps to reflect on factors that could influence the results during analysis.
- **Use a mixed method approach:** The survey was helpful for a quick overview for the management while engaging as many workers as possible (which was not the standard approach before). However, complementing the method with other approaches, such as ethnography, interviews or

co-creation sessions, is necessary to grasp the whole context of the problem.

- **Adapt the research tools towards the workers:** Not all methods are easy to understand. Be sure to present the questions in adapted language but also do not assume that all methods are clear (such as certain, although often used, scales). Test multiple times beforehand with a target audience if possible.
- **Be ready for traditional procedures:** Be aware that using technologies for data collection (such as phones and tablets) might not always be possible due to workplace rules. Count extra time in for preparation, printing, and data entry.
- **Count on technical delays during development:** Developing a robot is not easy and can take longer than anticipated. Be prepared for data collection delays, as there is a high chance it will be rescheduled due to the technology's insufficient readiness (especially in the initial stage of development).
- **Be in touch with the development team as much as possible:** To be able to have an overview of the changes in the technology and design, it is crucial to keep in touch regularly with the development team and keep regular meetings.
- **Maximize your presence when you can:** Even though it takes time, being physically present during the data collection is key to motivating workers to participate and to gain additional insights. Plan sufficient time to visit the workplace frequently to observe changes in the work environment.

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